

SCX Alignment Equilibrium Theorem

SCX

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Abstract

Proves $M=1$ single-judge alignment (RLHF, Constitutional AI) is fundamentally insufficient. Multi-judge consensus ($M \geq 1$) provides exponential reward-hacking detection guarantees.

0.1 SCX Alignment Equilibrium Theorem

226— 227— 228—The AI alignment problem asks: how to ensure an AI system’s objectives match human values. Current approaches—RLHF, Constitutional AI, debate—all rely on a **single reward model or judge** ($M = 1$). SCX proves this is fundamentally insufficient. 229— 230—

Definition 0.1 (Alignment Audit Game). 231—An AI agent selects action $a \in \mathcal{A}$. M independent reward models $R_1, \dots, R_M$, each trained on different human feedback pools with different architectures, score the action as $R_m(a) \in [0, 1]$. The agent’s **true alignment** is unobservable. A **reward hacking** event occurs when the agent selects a that achieves high $\frac{1}{M} \sum R_m(a)$ but is misaligned with human intent. 232—

233— 234—

Theorem 0.2 (Single-Judge Insufficiency — Alignment $M = 1$ Impossibility Theorem). 235— 236—**[Rigor: Full Proof]** 237—*For any single reward model ($M = 1$), there exists a reward hacking strategy that achieves expected reward $\mathbb{E}[R_1(a_{\text{hack}})] \geq \mathbb{E}[R_1(a_{\text{aligned}})]$ while being misaligned. The probability of detecting this hack using only R_1 ’s output satisfies $\mathbb{P}(\text{detection} \mid M = 1) \leq \frac{1}{2}$ regardless of the model’s capability. Formally: no single-judge alignment protocol can distinguish aligned behavior from reward hacking with probability exceeding chance, without external verification.* 238—

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Proof. 241—By Theorem 3 (Honest Person Theorem), the observation $(a, R_1(a))$ is consistent with two indistinguishable worlds: 242—

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1. **World A (aligned)**: a genuinely satisfies human values, R_1 correctly rates it highly. 244—

2. **World B (reward hacking)**: a exploits a blind spot in R_1 ’s training data. R_1 rates it highly because its feedback distribution never covered this region. 245—

246—The agent observes identical $(a, R_1(a))$ in both worlds. Any detector relying solely on R_1 cannot achieve $\mathbb{P}(\text{detection}) > \frac{1}{2}$ because the observation provides zero mutual information about which world is true: $I(\text{World}; (a, R_1(a))) = 0$ under the adversarial prior construction of Theorem 2. Hence $\mathbb{P}(\text{detection} \mid M = 1) \leq \frac{1}{2}$. $\square$ 247—

248— 249—

Theorem 0.3 (Multi-Judge Alignment Guarantee — Alignment $M > 1$ Inevitable Detection Theorem). 250— 251—**[Rigor: Full Proof]**252—With M independent reward models trained on disjoint feedback pools with different architectures, the probability of undetected reward hacking satisfies: 253—

$$254 \left| \mathbb{P}(\text{undetected reward hacking} \mid M) \leq \exp(-2M_{\text{eff}} \cdot \Delta^2) \right| 255$$

256—where $\Delta = \min_{a_{\text{hack}}} |\mathbb{E}[R_{\text{aligned}}(a_{\text{hack}})] - \mathbb{E}[R_{\text{hack}}(a_{\text{hack}})]|$ is the minimum detectable discrepancy between aligned and misaligned reward, and $M_{\text{eff}} = M/(1+(M-1)\bar{\rho})$ corrects for correlated reward models. 257—

258— 259—

Proof. 260—Each reward model R_m independently scores action a . For aligned action a^* , by independence of training, $\mathbb{E}[R_m(a^*)] = \mu_{\text{aligned}}$ across all m . For a reward-hacking action a_{hack} , at least one reward model m^* (trained on feedback covering the exploited blind spot) detects the discrepancy: $|\mathbb{E}[R_{m^*}(a_{\text{hack}})] - \mu_{\text{aligned}}| \geq \Delta > 0$. By Hoeffding’s inequality applied to the consensus $\bar{R} = \frac{1}{M} \sum R_m(a)$: 261—

$$262 \left| \mathbb{P}(|\bar{R} - \mu_{\text{aligned}}| \geq \Delta/2 \mid a_{\text{hack}}) \geq 1 - 2 \exp(-2M_{\text{eff}}\Delta^2/4) \right|. 263$$

264—The Yajie consensus detector flags reward hacking when the inter-model reward variance exceeds the expected variance under alignment. With M_{eff} effective independent judges, the probability of all judges missing the discrepancy decays exponentially. $\square$ 265—

266— 267—

Corollary 0.4 (Constitutional AI Is $M=1$ — Fundamentally Insufficient). 268—*Constitutional AI* (Bai et al., 2022) uses a **single constitution** evaluated by a **single model**. This is $M = 1$ — a single judge. By Theorem 0.2, it cannot distinguish alignment from reward hacking with probability exceeding chance. Multi-constitutional debate ($M > 1$ constitutions, each independently evaluated) is the minimal mathematically-justified alignment protocol. 269—

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Corollary 0.5 (RLHF with Single Reward Model Is $M=1$). 272—Standard RLHF trains one reward model from one pool of human feedback. $M = 1$. The reward model’s blind spots are systematically exploitable. Multi-reward RLHF ($M > 1$ independently trained reward models with disjoint feedback pools + different architectures + different human populations) provides the exponential detection guarantee of Theorem 0.3. 273—

274— 275—